

Test Evidence — Smart Scientific Calculator

1. Purpose and Method

This document records the evidence for the 19 test cases in the Test Cases document. On 3 October 2026 an automated script loaded the repository's real public/index.html and public/script.js, clicked the actual on-screen buttons, and sent requests to the repository's real server.js. The values below were captured from that run.

- **Real:** all front-end code, the Express server and its /api/history routes, and every button click.
- **Stand-in:** no MongoDB server was available in the test environment, so the database layer was replaced by an in-memory substitute that keeps the same save and find-latest-10 behaviour. Results for TC-016 to TC-019 should be re-confirmed once against a real MongoDB instance.
- **Screenshot:** Figure 1 is a real screenshot of the running application taken by the team.

2. Application UI

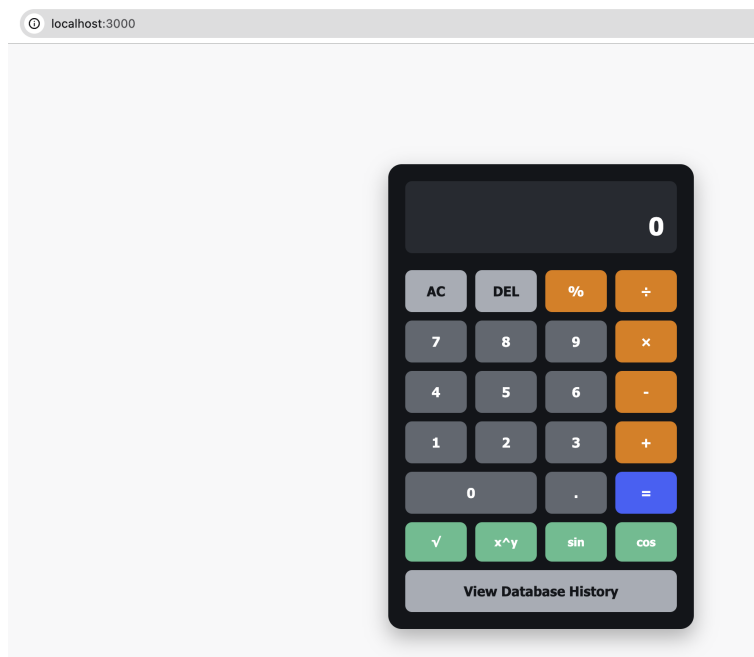


Figure 1: Application running at localhost:3000 (initial state, display 0). It shows the Iteration 1 keypad (AC, DEL, %, ÷, ×, -, +, digits, =), the Iteration 2 & 3 scientific row (√, x^y, sin, cos), the Iteration 5 grid layout with display screen, and the Iteration 6 “View Database History” button.

3. Evidence by Test Case

ID	Input sequence → display after each click	Observed result	Status
TC-001	5→"5" +→"5 + " 3→"5 + 3" ⇒"8"	Display shows 8	Pass
TC-002	2→"2" -→"2 - " 5→"2 - 5" ⇒"-3"	Display shows -3	Pass
TC-003	2→"2" .→"2." 5→"2.5" ×→"2.5 * " 2→"2.5 * 2" .→"2.5 * 2." 5→"2.5 * 2.5" ⇒"6.25"	Display shows 6.25	Pass
TC-004	4→"4" 5→"45" %→"45 % " ⇒"Error"	Display shows Error	Fail
TC-005	1→"1" 2→"12" 3→"123" AC→"0"	Display shows 0; history line ""	Pass
TC-006	1→"1" 2→"12" 3→"123" DEL→"12"	Display shows 12	Pass
TC-007	1→"1" 4→"14" 4→"144" √→"12"	Display shows 12	Pass
TC-008	3→"3" 0→"30" sin→"0.49999999999999994"	Display shows 0.49999999999999994	Pass
TC-009	6→"6" 0→"60" cos→"0.50000000000000001"	Display shows 0.50000000000000001	Pass

ID	Input sequence → display after each click	Observed result	Status
TC-010	2→"2" x^y→"1024"	Display shows 1024	Pass
TC-011	5→"5" ÷→"5 / " 0→"5 / 0" ⇒"Error"	Display shows Error	Pass
TC-012	-→" - " 9→" - 9" √→"Error"	Display shows Error	Pass
TC-013	+→" + " +→" + " ⇒"Error"	Display shows Error	Pass
TC-014	Clicked: .btn:active { transform: scale(0.95); }, .btn { transition: background-color 0.2s; }	Stylesheet defines .btn:active { transform: scale(0.95) } and a 0.2 s background-color transition	Pass
TC-015	1→"1" 2→"12" 3→"123"	Display shows 1, then 12, then 123	Pass

Note: a trailing space after an operator (for example "5 + ") is part of the expression text the application builds.

Iteration 6 — History & Database

TC-016 — request and response captured when 7 + 3 = was pressed:

POST /api/history

Body: {"expression":"7 + 3","result":"10"}

Status: 201 Created

Response: {"message": "Saved successfully", "record": {"expression": "7 + 3", "result": "10", "_id": "id9", "createdAt": "2026-10-03T18:04:55.716Z"}}

TC-017 — alert shown after 12 records were saved (only the 10 newest are listed, newest first):

Recent Database History:

1. 11 + 11 = 22
2. 10 + 10 = 20
3. 9 + 9 = 18
4. 8 + 8 = 16
5. 7 + 7 = 14
6. 6 + 6 = 12
7. 5 + 5 = 10
8. 4 + 4 = 8
9. 3 + 3 = 6
10. 2 + 2 = 4

TC-018 — alert with an empty database:

No history found in database.

TC-019 — alert with the server stopped:

Could not connect to database server.

4. Failed Test — TC-004 (DEF-01)

Steps from the test case (4, 5, %, =):

4→"4" 5→"45" %→"45 % " ⇒"Error"

Result: Error (expected 0.45) → Fail

Supplementary run showing what the % key actually does (1, 0, %, 3, =):

1→"1" 0→"10" %→"10 % " 3→"10 % 3" ⇒"1"

Result: 1 → % behaves as modulo (10 % 3 = 1)

5. Additional Observation

OBS-1. With the server stopped, a normal calculation (2 + 2 =) shows Error instead of 4, because the history request is made inside the same try block as the calculation. This is outside the 19 test cases but is worth fixing, since the calculator should still work when the database is unavailable.

2→"2" +→"2 + " 2→"2 + 2" ⇒"Error"

6. Execution Log

Run date : 2026-10-03 (UTC)

Node.js : v22.22.2 Express: 4.22.3 jsdom: 30.1.1

Server log: Server running on http://localhost:3000 | MongoDB connected successfully

TC-001 Pass Display shows 8

TC-002 Pass Display shows -3

TC-003 Pass Display shows 6.25

TC-004 Fail Display shows Error

TC-005 Pass Display shows 0; history line ""
TC-006 Pass Display shows 12
TC-007 Pass Display shows 12
TC-008 Pass Display shows 0.49999999999999994
TC-009 Pass Display shows 0.50000000000000001
TC-010 Pass Display shows 1024
TC-011 Pass Display shows Error
TC-012 Pass Display shows Error
TC-013 Pass Display shows Error
TC-014 Pass Stylesheet defines .btn:active { transform: scale(0.95) } and a 0.2 s background
TC-015 Pass Display shows 1, then 12, then 123
TC-016 Pass 201 Created; expression "7 + 3", result "10"
TC-017 Pass Alert lists 10 records; first entry "1. 11 + 11 = 22"
TC-018 Pass Alert "No history found in database."
TC-019 Pass Alert "Could not connect to database server."

TOTAL: 18 passed, 1 failed (TC-004), 19 executed

7. Evidence Summary

Metric	Count
Total cases	19
Passed	18
Failed	1 (TC-004)